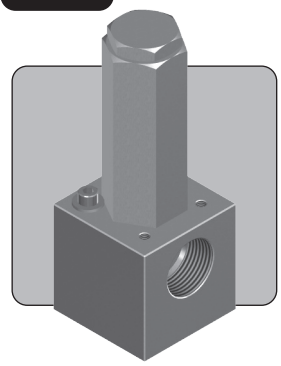
RL100

RELIEF VALVE

1″

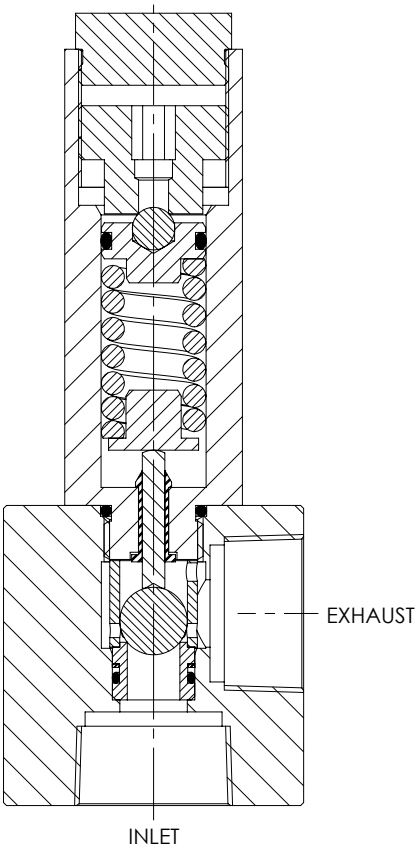
UP TO 207 BAR 3,000 PSI



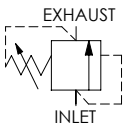
The RL100 is a 1″ ported relief valve. Designed for emergency relief applications, it may be used to protect systems up to 207 bar (3,000 psi).

The valve is a proportional metering style relief valve, and is approved as a safety accessory to category IV of the Pressure Equipment Directive, enabling its use in the most demanding applications.

- Approved to category IV of the Pressure Equipment Directive (2014/68/EU)
- Available preset via internal adjustment or with an external handwheel adjustment option
- Wire locking and tagging available on request
- Minimal leakage at 90% of set pressure
- Reseat within 20% of set pressure
- Repeatabily within ±5% of set pressure
- Subsea variants available, contact us for advice
- Suitable for use with mineral oils, water glycols and plain water
- Suitable for many other media, contact us for advice
- Various spring ranges available



Specifications

BASIC MODEL NUMBER	RL100
SYMBOL	
MAX WORKING PRESSURE	207 bar (3,000 psi)
MAX OUTLET PORT PRESSURE	69 bar (1,000 psi) Note: Any outlet port pressure is directly additive to the set pressure
ORIFICE SIZE	Ø11.1mm (Ø0.437″)
FLUID	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1″
WEIGHT	2.5 kg (5.5 lb)

Specifications may change without notice

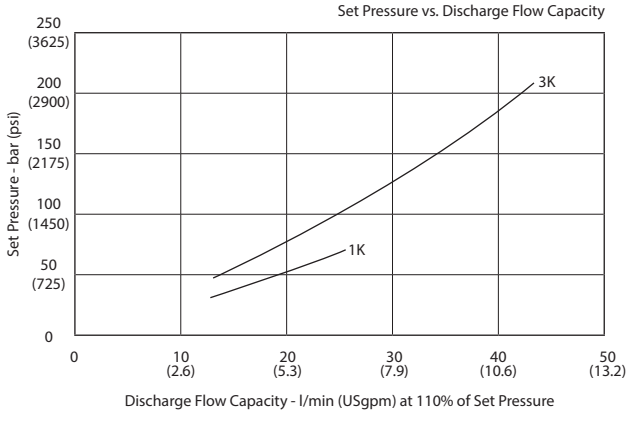
Materials

Externally Exposed Parts: 316 stainless steel on all pressure retaining components, and 300 series stainless steels on non-critical parts.

Internally Wetted Parts: 316 and 302 stainless steel, silicon nitride and acetal. The standard valve is designed for use with mineral oil, water glycols and plain water. Contact us for advice when use with other media is required. Note that use with water above 100°C is not recommended.

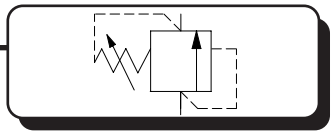
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

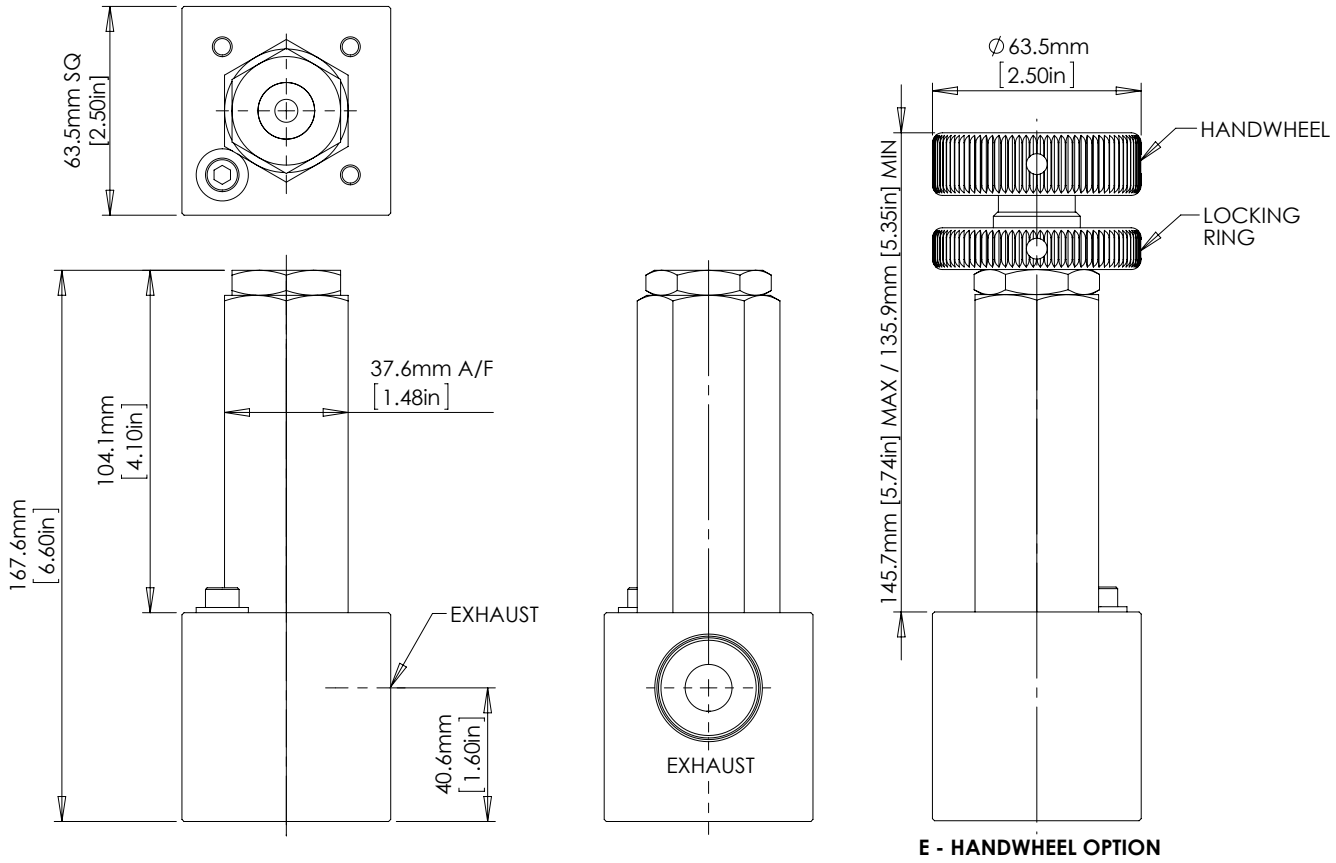
Typical Performance



Typical performance based on water
Contact us with system details for an application specific calculation

Installation Diagram





E - HANDWHEEL OPTION

Product Selector

RL100 P - E - NBR - 1.6K

VALVE TYPE	PORTING OPTIONS	VALVE OPTIONS	SEAL OPTIONS	SET PRESSURE RANGE
RL100	P 1" BSPP female inlet and exhaust N 1" NPT female inlet and exhaust	Leave blank if none required E Handwheel (external) adjustment ⚠ Exposure to high and low fluid temperatures will influence handle temperature. Operator safety must be considered	Leave blank for Viton® seals -10˚C to +120˚C NBR Nitrile seals -30˚C to +105˚C EP EPDM seals -40˚C to +120˚C KLZ Kalrez® seals -25˚C to +120˚C HNBR Hydrogenated Nitrile seals -20˚C to +120˚C Further seal options are available on request	1K 28 - 69 bar (400 - 1,000 psi) 3K 48 - 207 bar (700 - 3,000 psi) Advise us of set pressure if factory pre-setting is required

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK